

XIANGYUE HUANG

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Summary

My research focuses on **efficient multi-GPU architectural support for large-scale AI systems**, leveraging workload characteristics and hardware behavior to improve scalability, memory efficiency, and performance. My work spans GPU and memory system architecture, with publications at ISCA (2) and MICRO (1).

Education

University of California, Santa Cruz

Ph.D. in Computer Science

Sept. 2024 - Present

Santa Cruz, USA

Wuhan University

B.E. in Computer Science

Sept. 2020 - June. 2024

Wuhan, China

Selected Projects

LIBRA: Efficient Multi-GPU Page Prefetcher

Sept. 2024 – April 2025

Multi-GPU System, Memory Management

- Designed and implemented a cost-aware and coordinated page prefetching system for UVM-based multi-GPU environments.
- Developed a multi-way, multi-stride prefetcher to accurately capture fine-grained memory access patterns within GPUs.

CDFD: Duplication-Centric Multi-GPU Memory Management

April 2025 – November 2025

Multi-GPU System, NVLink

- Developed a hybrid mechanism combining coarse-grained duplication with fine-grained deduplication to balance performance and overhead.
- Conducted detailed characterization of modern interconnects, revealing inefficiencies in traditional fine-grained data movement strategies.

PUBLICATIONS

- **Xiangyue Huang**, Yanan Guo, Yuanchao Xu, “LIBRA: A High-Accuracy, Cost-Aware, and Coordinated Multi-GPU Page Prefetcher”, The 53rd International Symposium on Computer Architecture. (**ISCA 2026**)
- **Xiangyue Huang**, Yanan Guo, Yuanchao Xu, “Coarse-Grained Duplication First, Fine-Grained Deduplication Later: Duplication-Centric Multi-GPU Memory Management”, The 53rd International Symposium on Computer Architecture. (**ISCA 2026**)
- Qizhong Wang, **Xiangyue Huang**, Yanan Guo, Yuanchao Xu, “Security and Performance Implications of GPU Cache Eviction Priority Hints”, The 58th IEEE/ACM International Symposium on Microarchitecture. (**MICRO 2025**)

Awards

- UCSC CSE Excellent PhD Award, 2026

Teaching Experience

- Teaching Assistant of CSE 130-02 - Principle of Computer System Design, UCSC, Fall 2024
- Teaching Assistant of CSE 13S-01 - Computer Systems and C Programming, UCSC, Spring 2026